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Course Code: A8753

VARDHAMAN
COLLEGE OF ENGINEERING

AUTONOMOUS

III B. Tech II Semester Assignment – I, February - 2026

(Regulations: VCE-R22)

CSE(AI&ML)

Date: 20-02-2026

Course Outcomes with Bloom's Levels:

CO#	CO Statement	Bloom's Level (L#)
CO1	Understand the fundamental concepts, evolution, and major categories of generative AI models.	L2
CO2	Analyze and apply transformer-based architectures for text generation, text style transfer, and creative AI-driven writing.	L3, L4
CO3	Design GAN-based models for image/video synthesis, translation, and artistic style generation.	L6
CO4	Apply AI techniques such as Music RNNs, the Musical Organ, and MuseGAN for automated music generation using structured music representations.	L3, L4
CO5	Evaluate and build multimodal generative AI systems by understanding architectures like DALL·E2 and Imagen.	L5, L6

Questions:

S No.	Question	Course Outcomes	Bloom's Levels
1	Summarize the architecture of Generative AI history and evolution in the context of automated legal contract analysis and summarization. Discuss the specific technical challenges and expected outcomes.	CO1	L2
2	Describe the role of GAN fundamentals and applications in the context of smart city traffic management and optimization. Discuss the specific technical challenges and expected outcomes.	CO1	L2

3	Demonstrate the implementation of Transformer Architectures in the context of cybersecurity threat simulation. Discuss the specific technical challenges and expected outcomes.	CO2	L3
4	Analyze the underlying mechanism of T5 models for text style transfer in the context of automated legal contract analysis and summarization. Discuss the specific technical challenges and expected outcomes.	CO2	L4
5	Formulate a novel pipeline integrating GAN architectures for video synthesis in the context of financial fraud detection systems. Discuss the specific technical challenges and expected outcomes.	CO3	L6
6	Propose an optimization strategy for GAN architectures for video synthesis in the context of climate modeling and weather prediction. Discuss the specific technical challenges and expected outcomes.	CO3	L6
7	Apply the principles of The Musical Organ architecture in the context of personalized e-learning and tutoring platforms. Discuss the specific technical challenges and expected outcomes.	CO4	L3
8	Compare and contrast the performance of MuseGAN for polyphonic generation in the context of robotic process automation and autonomous navigation. Discuss the specific technical challenges and expected outcomes.	CO4	L4
9	Evaluate the effectiveness of DrawBench human evaluation metrics in the context of smart city traffic management and optimization. Discuss the specific technical challenges and expected outcomes.	CO5	L5
10	Formulate an improved architecture using The Text Encoder (CLIP) and Prior in the context of automated legal contract analysis and summarization. Discuss the specific technical challenges and expected outcomes.	CO5	L6

Question 1

Core Concept

Generative AI refers to machine learning models capable of generating new content such as text, images, or music based on learned patterns from training data. The evolution of generative AI started with rule-based Natural Language Processing systems, followed by statistical models like n-grams. Later, deep learning approaches such as Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) improved sequence modelling.

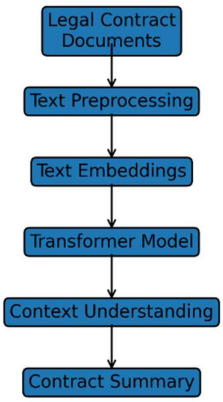
The most significant advancement came with **Transformer architectures**, which use self-attention mechanisms to understand relationships between words in large documents. Models such as GPT, BERT, and T5 can process long texts efficiently and generate meaningful summaries. In legal document processing, these models help automate contract analysis by extracting important clauses and generating concise summaries.

Contextual Application

Legal contracts are typically long and contain complex legal terminology. Manually reviewing these documents requires considerable time and expertise. Generative AI systems can automatically analyse legal contracts and produce summaries that highlight key information such as obligations, liabilities, payment terms, and termination clauses.

By applying transformer-based NLP models, organizations can automate contract review processes. This improves efficiency, reduces manual workload for legal professionals, and allows faster decision-making. The system can also classify contracts and detect potential risks in agreements.

Architectural Diagram



Technical Challenges

One of the main challenges is the complex structure of legal language, which often includes long sentences and domain-specific terminology. Another challenge is the large size of legal documents, which requires models capable of handling long contextual dependencies. Data privacy is also critical since legal contracts may contain confidential information. Additionally, training AI models with sufficient domain-specific legal data is necessary to ensure accurate results.

Expected Outcomes

The use of generative AI in legal contract analysis can significantly improve document processing efficiency. The system can automatically summarize lengthy contracts, highlight critical clauses, and assist legal professionals in reviewing documents faster. This technology can reduce human workload, improve accuracy in legal analysis, and support better decision-making in legal and business environments.

Question 2

Core Concept

Generative Adversarial Networks (GANs) are a type of generative AI model consisting of two neural networks: a Generator and a Discriminator. The generator creates synthetic data while the discriminator evaluates whether the generated data is real or fake. Through this adversarial training process, the generator gradually learns to produce highly realistic data.

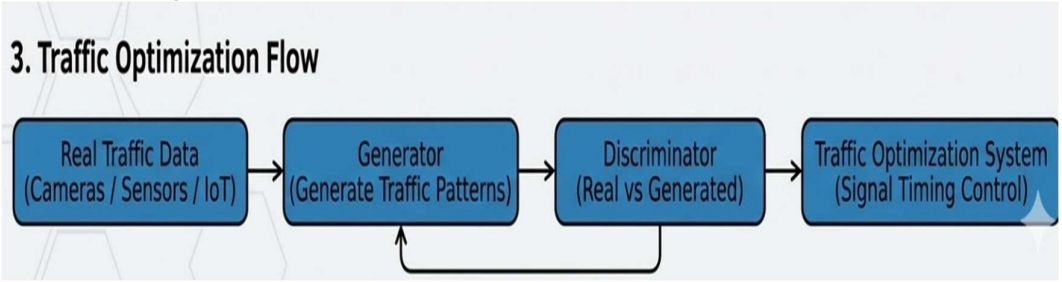
GANs are widely used in image generation, simulation, and data augmentation. In smart city systems, GANs can generate realistic traffic patterns. These generated patterns help urban planners and traffic management systems understand possible traffic situations and optimize signal control strategies.

Contextual Application

In smart cities, traffic management relies on large amounts of data collected from traffic cameras, road sensors, GPS devices, and IoT systems. GAN models can learn patterns from historical traffic data and generate realistic traffic flow simulations.

Using these simulations, traffic management systems can predict congestion, optimize signal timing, and improve road usage efficiency. GAN-based models can also simulate rare events such as accidents or heavy congestion, helping authorities test traffic control strategies in advance.

Architectural Diagram



Technical Challenges

One major challenge is the availability of high-quality traffic datasets, since accurate training requires large amounts of real traffic data. Another challenge is model stability, as GAN training can sometimes be unstable and difficult to optimize. Additionally, real-time traffic systems require fast processing and low latency, which can be computationally demanding. Integrating GAN models with existing smart city infrastructure and IoT systems is also technically complex.

Expected Outcomes

The application of GANs in traffic management can significantly improve smart city transportation systems. It can help predict congestion patterns, optimize traffic signal timings, and simulate various traffic scenarios for better planning. This leads to reduced traffic congestion, improved road safety, and more efficient urban transportation systems.

Question 3

Core Concept

Transformer architectures are advanced deep learning models designed for processing sequential data using a mechanism called self-attention. Unlike traditional RNN or LSTM models, transformers analyse relationships between all elements in a sequence simultaneously, allowing them to capture long-range dependencies efficiently.

In cybersecurity, transformers can analyse large volumes of network logs, system events, and security alerts. By learning patterns from historical attack data, transformer models can simulate potential cyber threats and detect unusual behaviour in network traffic.

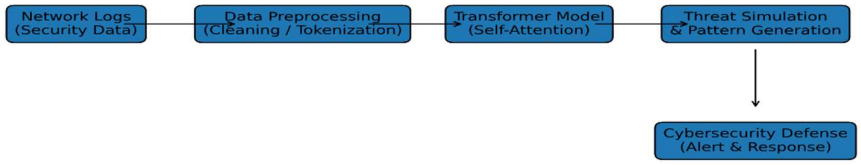
Contextual Application

Modern organizations generate huge amounts of cybersecurity data from firewalls, intrusion detection systems, network logs, and cloud services. Transformer models can process this data to identify hidden patterns related to cyber-attacks.

For example, a transformer-based system can learn patterns from previous attacks such as malware activity, phishing attempts, or unauthorized access attempts. Using this knowledge, the system can simulate possible future threats .

This approach improves threat prediction, anomaly detection, and automated cybersecurity monitoring.

Architectural Diagram



Technical Challenges

One major challenge is handling large-scale cybersecurity data, which requires high computational power and storage. Another difficulty is distinguishing between normal network activity and malicious behaviour, since many attacks mimic legitimate actions. Data privacy and security are also important concerns when training models on sensitive organizational data. Additionally, transformer models require large training datasets to achieve accurate threat detection.

Expected Outcomes

Implementing transformer architectures for cybersecurity threat simulation can improve the ability of organizations to detect and predict cyber-attacks. The system can automatically analyse network behaviour, simulate potential attack scenarios, and generate alerts for suspicious activities. This leads to stronger cybersecurity defences, faster threat response, and improved protection of digital infrastructure.

Question 4

Core Concept

The T5 (Text-to-Text Transfer Transformer) model is a transformer-based architecture developed to handle multiple NLP tasks using a unified text-to-text framework. In this approach, every task—such as translation, summarization, or classification—is treated as a text generation problem.

T5 uses an encoder–decoder architecture. The encoder reads and understands the input text, while the decoder generates the output text based on the encoded representation. This mechanism allows T5 to perform tasks like text summarization, style transfer, and information extraction effectively.

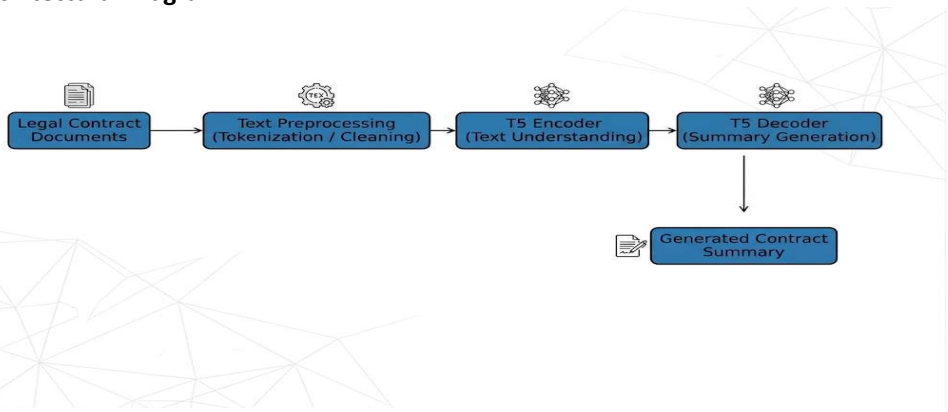
Contextual Application

In legal systems, contracts contain complex and lengthy clauses that are difficult for non-experts to understand. Using a T5-based model, the system can analyze legal contract documents and convert them into simplified summaries or structured explanations.

For example, the model can take a long legal paragraph and generate a concise summary explaining the main obligations, payment terms, and legal conditions. It can also perform text style transfer, converting highly technical legal language into simpler language that is easier to understand.

This improves efficiency for lawyers, businesses, and clients who need to quickly interpret legal documents.

Architectural Diagram



Technical Challenges

One challenge is the complex nature of legal language, which contains specialized terms and long sentence structures. Another difficulty is ensuring accurate summarization, since missing or misinterpreting legal clauses can lead to serious consequences. T5 models also require large datasets of legal documents for effective training. Additionally, protecting the privacy and confidentiality of legal data during training is important.

Expected Outcomes

The use of T5 models in legal contract analysis can automate document summarization and simplify complex legal language. This allows users to quickly understand key contract terms and reduces the time required for manual document review. As a result, legal professionals can improve productivity, reduce errors, and enhance decision-making in legal processes.

Question 5

Core Concept

Generative Adversarial Networks (GANs) consist of two neural networks: a Generator and a Discriminator. The generator creates synthetic data samples, while the discriminator evaluates whether the generated data is real or fake. Through this adversarial learning process, the model improves its ability to generate realistic data patterns.

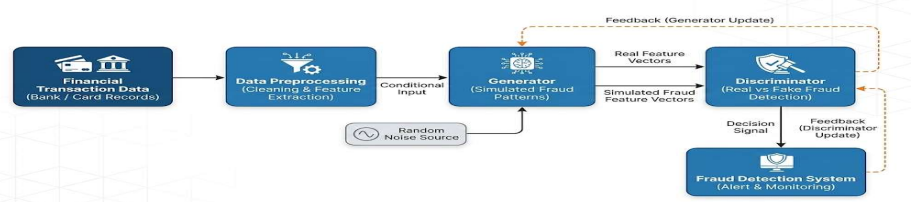
In financial systems, GANs can be used to simulate fraudulent transaction patterns and help train fraud detection systems. By generating synthetic fraud examples, GAN models can improve the ability of machine learning systems to detect unusual or suspicious financial activities.

Contextual Application

Financial institutions process millions of transactions daily, making manual fraud detection extremely difficult. GAN-based systems can analyse historical transaction data and generate realistic fraud scenarios such as unauthorized transactions, identity theft, or abnormal spending patterns.

These generated patterns help train fraud detection models to recognize suspicious activities more effectively. For example, GAN models can simulate new types of fraud that may not yet exist in historical datasets, allowing banks to prepare their systems in advance.

Architectural Diagram



Technical Challenges

One challenge is the imbalance between normal and fraudulent transactions, since fraud cases are relatively rare. Another issue is the quality of generated synthetic data, which must closely resemble real fraud patterns to be useful for training models. Additionally, financial data contains sensitive information, so maintaining data privacy and security is critical. GAN models can also be difficult to train and may require careful tuning to achieve stable performance.

Expected Outcomes

A GAN-based fraud detection system can improve the ability of financial institutions to identify suspicious transactions in real time. The system can simulate various fraud scenarios, enhance the training of detection models, and reduce financial losses caused by fraudulent activities. Ultimately, this technology helps improve security in digital banking and online payment systems.

Question 6

Core Concept

Generative Adversarial Networks (GANs) are deep learning models that consist of two components: a Generator and a Discriminator. The generator creates synthetic data samples while the discriminator evaluates whether the generated samples are real or artificial. Through continuous training, the generator learns to produce highly realistic data.

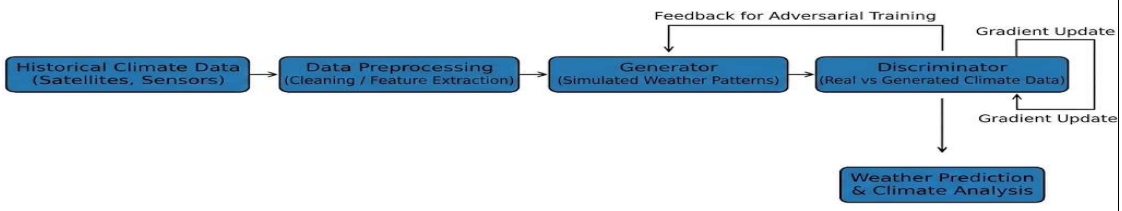
In climate science, GAN models can be used to generate realistic weather patterns and climate simulations based on historical environmental data. These generated patterns help researchers analyse future climate conditions and improve weather forecasting systems.

Contextual Application

Climate prediction relies on large datasets collected from satellites, weather stations, ocean sensors, and environmental monitoring systems. GAN models can learn complex patterns from this data and generate simulations of possible future weather conditions.

For example, GAN-based systems can simulate rainfall patterns, temperature variations, or extreme weather events such as storms and droughts. These simulations help meteorologists and climate researchers better understand environmental changes and make more accurate predictions about future climate trends.

Architectural Diagram



Technical Challenges

One challenge is the large volume and complexity of climate data, which requires significant computational resources for training models. Another difficulty is ensuring the accuracy of generated weather patterns, since incorrect predictions may affect critical decisions such as disaster management. Climate systems are also highly dynamic and influenced by many variables, making modelling more complex. Additionally, GAN models can sometimes face training instability, which must be carefully managed.

Expected Outcomes

Using GAN models for climate prediction can improve the ability to simulate complex weather patterns and environmental changes. These systems can support more accurate weather forecasting, better disaster management planning, and improved understanding of long-term climate trends. As a result, governments and organizations can make more informed decisions related to environmental sustainability and public safety.

Question 7

Core Concept

The Musical Organ model is a generative AI architecture designed to generate structured sequences by learning patterns from sequential data. It works similarly to sequence models such as RNN or LSTM, where the system learns relationships between elements in a sequence and generates new outputs based on learned patterns.

In generative AI, Musical Organ models are capable of generating structured outputs such as music or other sequential content. When applied to educational systems, the same concept can be used to generate personalized learning materials by analyzing student behavior and learning patterns.

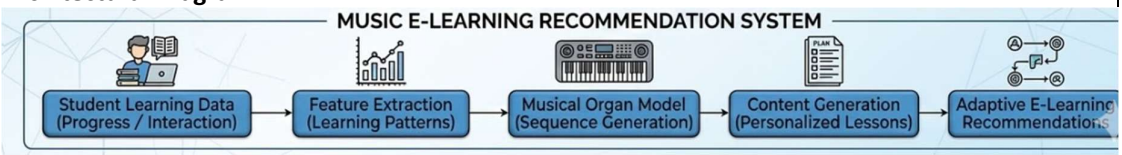
Contextual Application

In modern e-learning platforms, students interact with digital learning systems through quizzes, assignments, and online lessons. These platforms collect large amounts of student performance data, including learning speed, strengths, and weaknesses.

Using the Musical Organ architecture, the system can analyze this data and generate adaptive educational content tailored to individual students. For example, if a student struggles with a particular topic, the model can automatically generate additional practice exercises or simplified explanations.

This approach helps create personalized learning experiences, improving student engagement and learning outcomes.

Architectural Diagram



Technical Challenges

One challenge is collecting accurate and diverse student learning data for training the model. Another difficulty is ensuring that the generated educational content remains accurate and pedagogically meaningful. Additionally, e-learning systems must maintain data privacy and security, since student data is sensitive. Designing models that adapt to different learning styles can also be technically complex.

Expected Outcomes

The implementation of Musical Organ models in e-learning systems can significantly improve personalized education. The system can generate adaptive learning materials, recommend suitable content for each student, and support continuous learning improvement. This leads to better student engagement, improved academic performance, and more efficient digital education platforms.

Question 8

Core Concept

MuseGAN is a generative adversarial network designed specifically for polyphonic music generation. Unlike traditional music generation models that generate a single melody line, MuseGAN can generate multiple music tracks simultaneously, such as drums, bass, guitar, and piano.

The architecture consists of a Generator that creates multi-track music sequences and a Discriminator that evaluates whether the generated music is realistic compared to real music

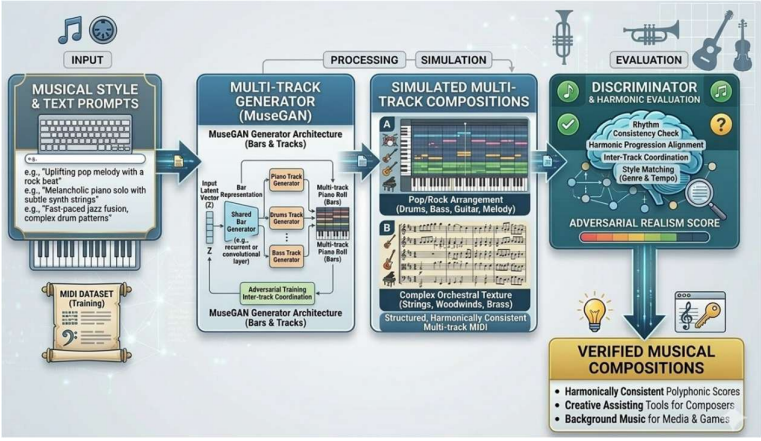
samples. Through adversarial training, the generator gradually learns to produce structured and harmonically consistent music.

Contextual Application

In automated music composition systems, MuseGAN can be used to generate complete musical arrangements rather than just single-note melodies. By training on large datasets of MIDI music files, the model learns relationships between different musical instruments and rhythm patterns.

For example, MuseGAN can automatically generate background music for games, films, or digital content. It can also assist musicians by producing new musical ideas and compositions. Compared to simple sequence models, MuseGAN provides better coordination between multiple instruments, making the generated music more realistic.

Architectural Diagram



Technical Challenges

One major challenge is maintaining musical harmony and synchronization between multiple tracks. Another issue is the large amount of training data required for learning complex musical structures. GAN models can also experience training instability, making it difficult to produce consistent music outputs. Additionally, ensuring diversity in generated compositions while maintaining musical quality can be challenging.

Expected Outcomes

Using MuseGAN for music generation can enable automated creation of complex musical compositions with multiple instrument tracks. The system can support creative music production, assist composers in generating new ideas, and provide background music for digital media applications. Overall, MuseGAN improves the realism and complexity of AI-generated music.

Question 9

Core Concept

DrawBench is a benchmark framework used to evaluate the performance of text-to-image generative models. It measures how accurately a generative AI model can convert textual prompts into meaningful visual outputs. The benchmark evaluates image quality, semantic accuracy, and alignment between the input text prompt and the generated image.

In multimodal generative AI systems, models process multiple data types such as text and images. DrawBench helps researchers test whether these models correctly interpret textual instructions and generate realistic visual scenarios.

Contextual Application

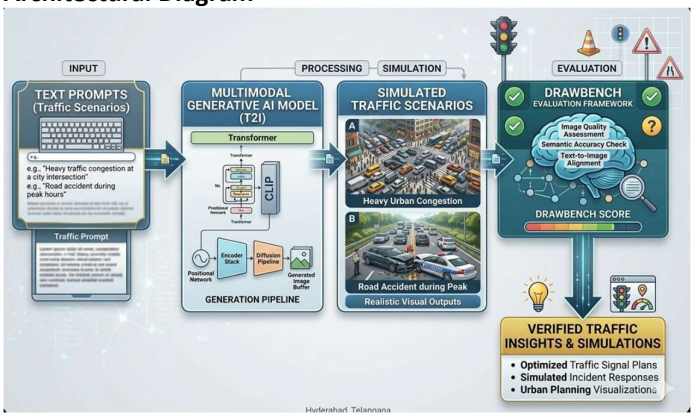
In smart traffic management systems, generative AI can be used to simulate different traffic scenarios using text prompts. For example, prompts like *"heavy traffic congestion at a city intersection"* or *"road accident during peak hours"* can be used to generate visual simulations.

Using DrawBench evaluation, the system can assess how accurately the generated images represent the intended traffic situation. These simulations can help city planners and transportation

authorities analyze possible road conditions, optimize traffic signals, and design better traffic control strategies.

This approach helps in urban planning, traffic simulation, and intelligent transportation system development.

Architectural Diagram



Technical Challenges

One challenge is ensuring that the generated images accurately represent complex traffic scenarios. Another difficulty is handling the large computational resources required for multimodal models. Additionally, evaluating visual outputs objectively can be challenging because different models may generate different interpretations of the same prompt. Maintaining real-time performance in traffic management systems is also an important challenge.

Expected Outcomes

Using DrawBench evaluation in traffic management systems can improve the reliability of generative AI models that simulate traffic conditions. These systems can help urban planners visualize possible traffic scenarios, test different traffic control strategies, and improve road infrastructure planning. Ultimately, this technology can contribute to safer and more efficient transportation systems.

Question 10

Core Concept

CLIP (Contrastive Language–Image Pretraining) is a multimodal AI model that learns relationships between text and other forms of data representations. It uses a text encoder and another encoder to map different data types into a shared representation space. A Prior model is then used to map textual information into a meaningful latent representation that can guide generation or analysis.

In generative AI systems, the combination of CLIP and Prior architectures allows models to understand complex textual descriptions and convert them into structured insights. This capability can be applied to analyze large textual datasets such as legal contracts.

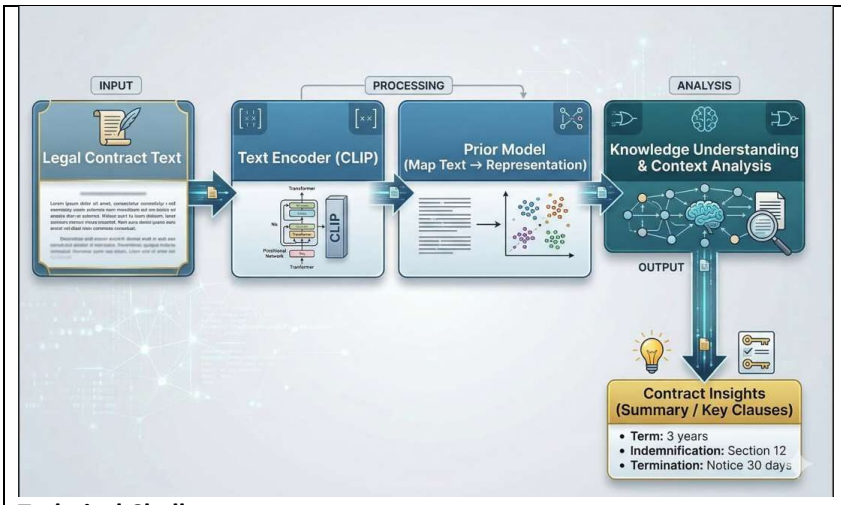
Contextual Application

Legal contracts often contain lengthy and complex clauses that require careful analysis. Using a CLIP + Prior architecture, the system can encode legal text into structured representations and identify relationships between clauses, obligations, and legal conditions.

For example, the system can analyze a contract and automatically highlight important sections such as payment terms, responsibilities, penalties, and termination clauses. It can also generate summaries that explain the key points of the agreement in a simplified manner.

This helps legal professionals and organizations quickly understand contract contents without manually reviewing every section.

Architectural Diagram



Technical Challenges

One challenge is handling the complex structure of legal documents, which often include technical terminology and long contextual dependencies. Another issue is ensuring the accuracy of extracted insights, since incorrect interpretation could lead to legal misunderstandings. Training multimodal models also requires large datasets and significant computational resources. Additionally, protecting confidential legal data during training and deployment is essential.

Expected Outcomes

The use of CLIP + Prior architectures can significantly improve automated legal contract analysis. These systems can quickly summarize large documents, identify key clauses, and provide structured insights from complex legal text. This improves efficiency in legal workflows, reduces manual effort, and supports faster decision-making in legal and business environments.